Intelligent Job Recommendation Browser Extension

For Job and Internship Recommendation for students and job seekers.

Prepared by

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April 2026

1. Problem Statement & Relevance

In recent times, students and early-career professionals face immense difficulty identifying job roles that genuinely align with their skills, projects, and academic background. The conventional job search forces candidates to manually browse multiple platforms, LinkedIn, Indeed, Naukri, Glassdoor repeatedly entering the same information while never feeling confident they have found the right fit.

This chaotic process creates a persistent fear of missing out (FOMO) on better opportunities, leaving candidates overwhelmed, exhausted, and misdirected at one of the most critical phase of their career.

The Core Gap

- Most portals rely on simple keyword-based filtering, ignoring depth of project experience and academic context.
- Students either overshoot applying for roles they are underqualified for or undersell themselves by targeting positions far below their potential.
- No widely available tool evaluates a candidate's complete profile holistically and delivers intelligent, ranked suggestions in real time.

Why it matters: Without an intelligent guidance system, misaligned applications, prolonged job searches, and decreased confidence become the norm for millions of graduates entering the workforce each year.

2. Proposed Solution

This project proposes an Intelligent Browser Extension that delivers real-time, personalized job role recommendations by automatically extracting and analyzing a candidate's profile with zero manual input required.

How It Works

1. **Data Extraction:** Automatically detects the active platform (LinkedIn, resume page, GitHub) and extracts technical skills, project descriptions, academic qualifications, and certifications via DOM parsing.
2. **Profile Processing:** Cleans, normalizes, and structures raw data into a standardized candidate profile JSON object using NLP preprocessing.
3. **Recommendation Engine:** Feeds the structured profile into a hybrid recommendation engine that scores and ranks job roles by match percentage, delivering top results with skill gap insights.

The extension operates as a lightweight popup seamlessly integrated into the user's browsing experience eliminating platform switching and repetitive data entry entirely.

3. Approach & Methodology

Stage 1 — Rule-Based Filtering

A predefined rule engine maps candidate attributes to eligible job role categories using logical criteria. If a candidate's skills include Python and their projects involve machine learning, the engine boosts the relevance score for roles like Data Scientist and ML Engineer. This stage efficiently narrows hundreds of roles to a relevant shortlist before ML processing begins.

Stage 2 — ML-Based Ranking (TF-IDF + Cosine Similarity)

Both the candidate profile and each job role description are converted into TF-IDF vectors. Cosine similarity is then computed between the candidate vector and each job vector to produce a relevance score. A weighted scoring model combines similarity with attribute-specific scores:

Attribute	Weight	Rationale
Technical Skills	40%	Primary industry filter
Project Experience	30%	Demonstrates practical ability
Academic Background	20%	Role eligibility baseline
Certifications	10%	Supplementary credibility

Skill Gap Analysis

After ranking, a differential analysis identifies missing skills between the candidate's profile and each recommended role's requirements. Each gap is mapped to curated learning resources, courses, certifications, and project ideas giving users a concrete improvement roadmap.

4. Tech Stack

Layer	Technology	Purpose
Extension UI	HTML, CSS, JavaScript (Manifest V3)	Popup interface & user interaction
Data Extraction	Content Script, DOMParser API	Profile extraction from web pages
Backend	Python + FastAPI	ML processing & REST API endpoint
Storage	Chrome Storage API / JSON	Job roles DB & local caching

5. Workflow & Architecture

The system follows a clean five-step pipeline from data extraction to recommendation display:

Extract: Content script activates on supported platforms, parses the DOM, and extracts skills, projects, education, and certifications.

Process: Background script receives the data, cleans and normalizes it, then sends it to the FastAPI backend via a REST API call.

Recommend: Backend applies TF-IDF vectorization, computes cosine similarity, applies the weighted scoring model, and runs skill gap analysis.

Return: Ranked recommendations, match percentages, missing skills, and improvement suggestions are sent back and stored in Chrome Storage.

Display: The popup UI renders an interactive dashboard with job role cards, match percentage bars, and skill gap details.

6. Key Features & Expected Outcome

Feature	Description
Ranked Job Recommendations	Top 10 job roles matched to candidate profile with a match % score
Automated Profile Extraction	Extracts skills, projects & education automatically no manual input

Skill Gap Analysis	Highlights missing skills per role with targeted learning suggestions
Multi-Platform Support	Works across LinkedIn, resume pages, and GitHub profiles
Real-Time Results	Delivers recommendations via browser popup

Expected Outcome

Drastically reduced time and effort spent searching across multiple job platforms.
Clear, data-driven understanding of which roles best match the candidate's current profile.
An actionable roadmap for bridging skill gaps and improving overall employability.
Reduced FOMO and increased confidence in career decision-making for students and graduates.

By bridging the gap between individual potential and industry demand, this system democratizes access to personalized career guidance making smarter, faster, and more confident career planning available to every student and job seeker, regardless of background or resources.